

NHA TRANG, Viet Nam

WMO: 488770

Lat: 12.228N

Lon: 109.192E

Elev: 10

StdP: 101.20

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.2	21.1	15.5	11.0	22.0	16.5	11.8	22.6	11.3	24.4	10.4	24.6	1.3	290	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	5.5	33.5	26.6	32.9	26.6	32.4	26.5	27.8	31.6	27.5	31.4	27.2	31.1	3.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.4	30.5	26.4	21.9	30.2	26.1	21.5	30.0	89.1	31.9	87.5	31.5	86.2	31.3	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.3	7.3	17.8	35.8	2.4	1.6	16.0	36.9	14.6	37.8	13.2	38.7	11.4	39.8		
(5)			15.8	28.7	1.7	0.5	14.6	29.0	13.5	29.3	12.6	29.6	11.3	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.4	24.5	25.1	26.4	28.1	29.1	29.4	29.2	29.2	28.5	27.3	26.3	25.1	
	DBStd	2.09	1.19	1.30	1.28	1.10	1.08	1.06	1.00	1.10	1.02	1.08	1.18	1.35	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	6336	449	422	509	543	592	582	595	597	555	537	488	467	
	CDD18.3	3294	191	188	251	293	333	332	337	338	305	278	238	209	
	CDH23.3	34199	1047	1261	2236	3280	4082	4171	4133	4178	3516	2837	2065	1394	
	CDH26.7	11467	64	152	480	1107	1711	1838	1736	1793	1312	783	363	128	
Wind	WSAvg	2.3	3.4	3.1	2.6	2.0	1.6	1.4	1.4	1.4	1.4	2.0	3.0	4.0	
Precipitation	PrecAvg	1673	47	10	36	68	109	71	48	87	218	364	404	211	
	PrecMax	2746	138	36	200	221	257	256	155	153	467	971	933	1269	
	PrecMin	963	10	0	0	19	33	29	0	29	83	171	94	6	
	PrecStd	462	37	11	61	51	61	50	34	32	93	169	231	249	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.9	29.9	31.2	32.6	33.9	34.4	33.9	34.2	33.2	31.7	30.4	29.5	
		MCWB	24.3	24.6	25.4	26.6	27.2	26.9	26.4	26.2	26.6	26.3	25.6	24.8	
	2%	DB	28.0	29.1	30.4	31.9	33.0	33.4	33.0	33.2	32.5	30.9	29.8	28.6	
		MCWB	23.8	24.4	25.2	26.3	27.0	26.7	26.4	26.4	26.6	26.0	25.5	24.5	
	5%	DB	27.4	28.3	29.7	31.3	32.4	32.7	32.3	32.5	31.9	30.4	29.2	28.0	
		MCWB	23.4	24.0	25.0	26.1	26.9	26.6	26.3	26.4	26.4	25.8	25.2	24.1	
	10%	DB	26.8	27.6	29.0	30.7	31.8	32.0	31.7	31.9	31.2	29.8	28.6	27.3	
		MCWB	23.0	23.7	24.7	26.0	26.8	26.5	26.2	26.3	26.2	25.7	25.0	23.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.2	25.6	26.3	27.6	28.2	28.2	27.9	28.1	27.8	27.2	26.7	26.0
			MCDB	27.7	28.5	29.6	31.1	31.9	32.0	32.0	31.9	31.6	30.4	29.1	28.2
2%		WB	24.7	25.0	26.0	27.1	27.7	27.7	27.3	27.5	27.3	26.7	26.2	25.3	
		MCDB	27.1	28.0	29.1	30.5	31.5	31.7	31.3	31.4	31.1	29.8	28.8	27.5	
5%		WB	24.1	24.6	25.6	26.7	27.3	27.3	26.9	27.1	26.9	26.4	25.7	24.9	
		MCDB	26.5	27.5	28.7	30.1	31.2	31.3	30.9	31.1	30.8	29.5	28.3	27.1	
10%		WB	23.6	24.2	25.2	26.4	27.0	26.9	26.6	26.7	26.6	26.0	25.4	24.4	
		MCDB	26.1	26.9	28.3	29.7	31.0	31.0	30.6	30.8	30.4	29.2	27.9	26.6	
Mean Daily Temperature Range		5% DB	MDBR	4.5	4.9	5.0	5.1	5.5	5.5	5.5	5.6	5.6	4.8	4.0	3.8
			MCDBR	5.1	5.4	5.6	5.6	6.0	6.1	6.0	6.2	6.3	5.5	5.0	4.7
	MCWBR		2.8	2.9	2.7	2.5	2.8	2.7	2.6	2.7	2.9	2.7	2.8	2.6	
	5% WB	MCDDBR	4.4	5.0	5.1	5.2	5.7	5.7	5.6	5.8	5.9	5.1	4.4	3.8	
		MCWBR	2.8	2.9	2.7	2.6	3.0	2.9	2.8	2.9	3.1	2.7	2.7	2.4	
Clear-Sky Solar Irradiance	taub	0.456	0.446	0.488	0.470	0.450	0.436	0.430	0.432	0.445	0.487	0.478	0.478	0.491	
	taud	2.283	2.309	2.204	2.285	2.374	2.437	2.440	2.423	2.380	2.245	2.259	2.259	2.174	
	Ebn at Noon	846	871	839	846	849	851	859	868	864	823	819	801	801	
	Edh at Noon	132	133	150	138	124	115	115	119	124	140	134	144	144	
All-Sky Solar Radiation	RadAvg	4.26	5.31	5.66	6.17	5.86	5.52	5.41	5.39	5.15	4.42	3.94	3.94	3.56	
	RadStd	0.45	0.46	0.53	0.32	0.41	0.31	0.35	0.41	0.42	0.46	0.54	0.54	0.60	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.38	+0.66	N/A	N/A	+0.34	+0.32	N/A	N/A	N/A
(47) Regional (3 neighbors)		+0.25	+0.47	N/A	N/A	+0.27	+0.28	N/A	N/A	N/A

Nomenclature: See separate page